WATCHING: BIOPLASTICS

A SERIES ON ECONOMIC-DEMOGRAPHIC-CONSUMER & TECHNOLOGY TRENDS IN SPECIFIC PLASTICS END MARKETS

Spring 2018

PLASTICS INDUSTRY ASSOCIATION

BETTER INDUSTRY. BETTER WORLD.

Spring 2018

The Plastics Industry Association (PLASTICS) sends special thanks to PLASTICS' Bioplastics Division, Brand Owners Council, and Transportation and Industrial Plastics Committee for their guidance and input on this Bioplastics Market Watch Report.

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Plastics Market Watch

Watching: Bioplastics

A series on economic-demographic-consumer & technology trends in specific plastics end markets

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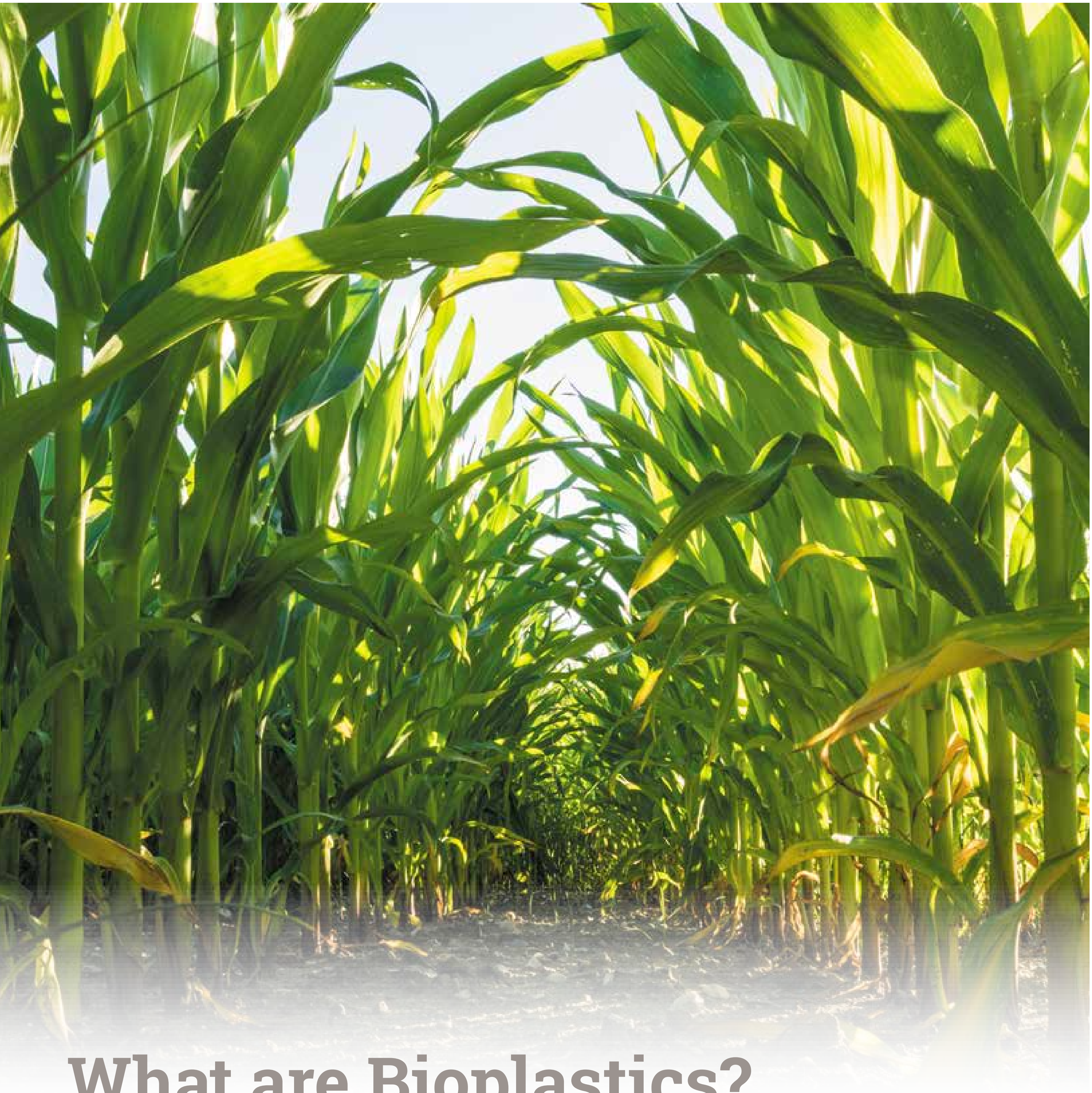


What are Bioplastics?

What are Bioplastics?

BI·O·PLAS·TIC

/ˌbīō ˈplastik/ noun
plural noun: bioplastics

1. a type of plastic partially or fully biobased and/or biodegradable
   a. a biobased bioplastic has some or all of its content produced from a renewable plant (or sometimes animal) source.
   b. biodegradable plastics are those that completely degrade into carbon dioxide (CO_2), methane (CH_4), water (H20), and biomass through biological action in a defined environment and in a defined timescale.
      i. Examples of types of biodegradability include compostable, anaerobically digestible, and marine and soil biodegradable.

For a product that has been around for 155 years, the understanding of bioplastics continues to be inconsistent, but it is becoming a more common tool in the plastics toolbox and in a wider range of industries.

Is it biodegradable? What is it made of? How do they perform? Is it recyclable? How can it be used? Can I compost it? Are they good for the environment? Do they impact food production?

The Plastic Industry Association's Bioplastics Division was established in 2007 and today has 22 members around world. Members include leading developers and manufacturers of bioplastics as well as members who represent leading sources for bioplastics, as well as brands that utilize the material in products and packaging. The Bioplastics Division wants to educate colleagues in the plastics sector as well as brands, consumers and policymakers about bioplastics and articulate the different bioplastics options available today and in the future.

PLASTICS published a bioplastics Market Watch Report in the summer of 2016. It was a valuable update on the status of bioplastics in the marketplace as well as research labs around the country. The report was useful with a number of key stakeholders in the industry. However, 24 months is a lifecycle of innovation and advancements in the area of bioplastics.

Industry understanding of bioplastics is advancing significantly, particularly in markets like Europe where government mandates are increasingly being used to address environmental issues and individual brands advance environmental sustainability initiatives that embrace bioplastics. And as consumers interact with bioplastics, their understanding increases as well.

In a survey PLASTICS conducted of 1,000 U.S. consumers in January 2018, 31 percent were still not familiar at all with bioplastics—defined as a type of plastic made from biobased materials like sugar cane or cornstarch, or are capable of biodegrading. At the other end of the spectrum, 32 percent were "familiar" or "somewhat familiar" with bioplastics.

While the numbers may seem surprising—particularly for the industry that knows the history and capabilities of bioplastics—they are an improvement from a similar PLASTICS survey conducted in 2016 when only 27 percent of U.S. consumers were somewhat or very familiar with bioplastics and 34 percent were not familiar at all with bioplastics.

The confusion among consumers may not be just with bioplastics, but with a wide range of products that fall under the "green" umbrella. "It's hard to know which is the truly green option in any product on the supermarket shelf. Is it better if the packaging is fully or partially biobased, is recyclable in theory or in practice or is capable of being composted or is actually composted, or indeed not packed at all?" explained Derek Atkinson, senior business director, The Americas, for Total Corbion PLA, a Netherlands-headquartered company. The reality is, all of these factors should be taken into account and overall environmental benefits determined through life cycle analysis, which considers material selection, product design, use and end of life disposition. Considering and evaluating all of those factors can reveal where the biggest environmental gains can be made.

The 2018 PLASTICS survey indicated that about 37 percent have seen or used a product made from bioplastic (biobased and/or compostable), an indication that many consumers may very well have touched or used a bioplastic product without knowing its origin or composition. PLASTICS Chief Economist Perc Pineda, who conducted the 2018 consumer survey noted, "A high percentage of surveyed respondents believe they have not seen or used a product made from bioplastic—either biobased or biodegradable. Continuing to educate consumers on bioplastics would go a long way."

PLASTICS Bioplastics Division

- BASF Corporation
- BiologiQ Inc.
- Braskem America
- Center for Bioplastics and Biocomposites
- The Coca-Cola Company
- Danimer Scientific
- Deere & Company
- DowDuPont, Inc.
- Earth Renewable Technologies
- Eastman Chemical Company
- Genarex LLC
- Heritage Bags™
- Jarden Plastic Solutions
- JinHui ZhaoLong High Technology Co.
- Leistritz
- NatureWorks LLC
- Novamont North America, Inc.
- PepsiCo
- Plastic Technologies, Inc.
- PolyOne Corporation
- Teknor Apex
- Total Corbion PLA

The array of bioplastics has expanded over the years—and by most indications will continue to diversify.

biobased

biodegradable or

+ both biobased and biodegradable.

Bioplastics are defined as a plastic partially or fully biobased and/or biodegradable. The first bioplastics were developed from traditional agricultural and renewable resources such as corn, sugar cane and soybeans. Second-generation bioplastics, and many being developed and introduced in the marketplace today, come from non-food renewable sources including food byproducts, wood and sawdust. The next generation of bioplastics—found in labs and trial testing—come from algae and other organisms away from food production that minimize environmental impacts.

There are 21 bioplastic polymers in the marketplace—with many more being developed and researched.

Articulating the definition of bioplastics, particularly when traditional fossil-based plastics can have bioplastic characteristics like biodegradability can be confusing. Adam Gendell, associate director of GreenBlue's Sustainable Packaging Coalition wrote in a Packaging Digest article, "To unpack this, that means that a bioplastic can be inherently non-biodegradable. It means that a bioplastic can contain zero percent biobased materials. A bioplastic may be 100 percent fossil-based. It can be any combination of being partially biobased, fully biobased, non-biobased, biodegradable, compostable or non-biodegradable, so long as it is not both non-biobased and non-biodegradable."

Gendell's article was entitled, "It's time for bioplastics to be plastics," which is a sentiment many have been supporting. NatureWork's Steve Davies, director of public affairs and communications, has advocated that calling plastic packaging by any other name, like bioplastic, is counterproductive and that it should be called plastic, "I wish people would stop calling PLA a bioplastic and start thinking of it as one more functional material—dare I say it…another plastic."

Davies, looking toward the packaging sector, believes the word "plastic" conjures up an array of polymers with distinctive performance characteristics, price points, sustainability qualities, and overall advantages and disadvantages to other packaging materials. "Bioplastic" as a word according to Davies carries vague or misunderstood connotations including compostability, bio-make up, limits to overall functionality, and higher costs.

	Non-biodegradable	Biodegradable
Biobased	BIOPLASTIC	BIOPLASTIC
Fossil-based	PLASTIC	BIOPLASTIC

PLASTICS uses the technical definition of bioplastic – a plastic that is biobased, biodegradable, or both biobased and biodegradable – in determining membership in the Bioplastics Division. In this report, the term biodegradable is used to discuss a polymer, or is a quote/reference from a third party. Finished products that make claims to a form of biodegradability (home/industrial compostable, marine/soil biodegradable, and anaerobic digestible) must be supported by a relevant standard, such as ASTM D6400 "Standard Specification for Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities." Biobased bioplastics can be supported by standards such as ASTM D6866 "Standard Test Methods for Determining the Biobased Content of Solid, Liquid, and Gaseous Samples Using Radiocarbon Analysis." Degradable plastics differ from biodegradable plastics, and are not considered bioplastics. Additional resources on supporting and making environmental claims can be found on the bioplastics website plasticsindustry.org/bioplastics or in the FTC Green Guides.

Bioplastic Polymers

Polymer Abbreviation	Polymer Name
PHA	Polyhydroxy Alkanoate
PLA	Polylactic Acid
TPS	Thermoplastic Starch
PBS	Polybutylene Succinate
PBAT	Polybutylene Adipate-Co-Terephthalate
PBAS	Polybutylene Adipate-Co-Succinate
PES	Polyethylene Succinate
PEF	Polyethylene Furanoate
PET	Polyethylene Terephthalate
PEET	Polyetherester Terephthalate
PTT	Polytrimethlene Terephthalate
PPA	Polyphthalamide
PA 410	Polyamide 410
PA 610	Polyamide 610
PA 1010	Polyamide 1010
PA 10	Polyamide 10
PA 11	Polyamide 11
TPC-ET	Thermoplastic Copolymer Elastomer
TPU	Thermoplastic Polyurethane
PE	Polyethylene
PP	Polypropylene
CA	Cellulose Acetate

PLASTICS, as an industry organization, is resin neutral and believes there is a growing need for a variety of plastics given the unique advantages plastics provide consumers, manufacturers and brand owners. Further, there is no polymer suitable for every need—and the best plastic is one that meets the functional needs of a given application or specification whether heat-resistant, barrier, or formable.

"So rather than the binary distinction between plastics and bioplastics, let's simply expand the list of functional characteristics on the polymer specification sheet. Buyers of plastics should continue to choose resins for their combination of functional material characteristics and cost performance," BlueGreen's Gendell concluded in an article.

Patrick Krieger, Assistant Director, Regulatory and Technical Affairs, PLASTICS

A Conversation with Patrick Krieger, PLASTICS Expert on Bioplastics

The Bioplastics Division at PLASTICS and other industry groups make a concerted effort to improve the understanding of bioplastics and get the information out to customers and consumers. Increased usage of bioplastics will be connected to a better understanding of what plastics and bioplastics are—and aren't—and how they are developed and can uniquely benefit an array of businesses and applications using plastics.

Q: What are bioplastics?

A: Besides cool? Bioplastics plastics are biobased, biodegradable, or both biobased and biodegradable. I know that definition sounds like an LSAT logic problem, but simply—if it's plastic and either comes from the earth or goes back to it, it's a bioplastic.

Biobased means the plastics are made at least partly from renewable resources, such as corn, sugar cane, soybeans, and agricultural byproducts like sawdust. And biodegradable means that it will completely break through natural processes within a short period of time.

A common misunderstanding about bioplastics is that "biobased" and "biodegradable" are linked; they are not. A bioplastic that is biobased may not necessarily be biodegradable, and a biodegradable bioplastic may not be biobased.

Q: Do people get confused about bioplastics?

A: Quite often. People think bioplastics are new, but they have been around for a long, long, time. The first natural plastics we used—rubber, horn, tortoise shells—were bioplastics. The first man-made plastic was also biobased, using cellulose from cotton. Henry Ford worked to develop plastics using renewable resources.

People think that polymers from a biopolymer feedstock will behave different than the petroleum feedstock. Biobased PE and PET are just as durable and just as recyclable.

People also think that biodegradable bioplastics will crumble into dust on their shelves. Biodegradability is very context specific—that's why we use terms like "industrial compostable" or "marine biodegradable."

I think this often comes down to awareness. People use bioplastics all the time, and they don't know it. That isn't a problem unique to PLA or PHA—stop a person on the street and ask them what a soft drink bottle is made out of, or what their cell phone is made from. Good design and performance is invisible, and often that means we are the victims of our own success.

Q: What are the environmental benefits of bioplastics?

A: At the beginning-of-life, bioplastics are made at least partly from sustainable resources (plants and other organic materials) and not fossil fuel (natural gas/petroleum). This can result in reduced carbon emissions. For example, Braskem estimates that 3.09 kilograms of CO_2 are removed from the air with each kilogram of their I'mGreen PE. At the end-of-life, the biodegradability and recyclability of bioplastics help reduce landfill usage and litter; these are some of the environmental benefits with bioplastics.

Q: How long does it take for bioplastics to biodegrade or compost?

A: Biodegradation rates depend upon a number of factors including the kind of bioplastic used in an application and the product's thickness. While logs and sticks in a forest are both "biodegradable," it will take the log a lot longer than the tiny stick to decompose. And the environment in which the product is degrading is important too. We differentiate between "home" and "industrial" composability because the latter is much more actively managed than the former, and as a result the temperatures get much higher.

It is important to remember, that all bioplastics need to be properly disposed—whether they are composted or recycled to another use. Just because something biodegrades doesn't mean it should be littered—you can get a ticket for tossing an apple core out the window just as easily as you can a cup.

Q: What is the most commonly used bioplastic?

A: There are 21 bioplastic polymers currently being used in the marketplace or under development—and more are being researched. Biobased Polyurethane (PUR) foams are the largest individual category of bioplastics used in the world. These are commonly used in seat cushions, seat backs and headrests. PET (Polyethylene Terephthalate) is the second most commonly produced bioplastic around the world; it is frequently used for bottles, packaging, and some fabric applications.

Q: Are bioplastics recyclable?

A: It varies based on the polymer, modifications, and format. Recyclability of bioplastics is similar to petroleum-based plastics in that bioplastics from biobased polymers without fillers are the easiest and most likely to be recycled. Bioplastics produced from polymer blends or that include biobased fillers may be difficult to recycle or may contaminate the existing recycling stream. End-of-life management of bioplastics is improving.

Q: Wouldn't the corn, soybean or other plants used for bioplastics be better used to feed people?

A: Not really, and for a lot of reasons.

Some people think that by diverting arable land from food to bioplastics, we inadvertently increase the cost of food. But bioplastics' use of agricultural feedstocks is minimal—0.067 percent of land according to European Bioplastics is used for bioplastics. By comparison, the USDA estimates that Americans waste 30 to 40 percent of our food supply.

> "Bioplastics are growing in those areas where they provide the most benefits for brands and consumers—whether it is a stadium looking to reduce its carbon footprint or an apparel designer filling a clothing niche."

From an environmental impact, producers of bioplastics are increasingly focused on how crops are grown, and are working with organizations like Bonsucro and the World Wildlife Fund to ensure that their feedstocks are produced sustainably. Braskem's Supplier Code of Conduct even goes beyond focusing on water, fertilizer, and pesticide use and mandates gender equality by producers.

And while there really is no food vs. bioplastics issue, there continues to be research into developing new sources from food waste, biomass and algae. So increasingly, this will be even less of an issue. Further, bioplastics use regionally-sourced materials in their development so a regionally abundant resource, like corn in the Midwest or agave in Mexico, can be processed into bioplastic materials.

Q: What are the most common applications for bioplastics?

A: Bioplastics—like all plastics—fulfill a wide (and growing) range of needs in many industries. The most common application for bioplastics is bottling and packaging. But you will also find bioplastics being used in textiles, food service packaging and cutlery, agriculture and horticulture, automobiles, building and construction, electronics, and consumer durables. Bioplastics are growing in those areas where they provide the most benefits for brands and consumers—whether it is a stadium looking to reduce its carbon footprint or an apparel designer filling a clothing niche.

Q: What are the driving forces behind the growth of bioplastics?

A: I think the underlying factors are the influence of end-of-life and beginning-of-life benefits to sustainability developed products, and how governments, corporations, and consumers have all begun to value sustainability more. But sustainability is not the sole requirement in polymer selection. If bioplastics weren't performing as they needed to, they wouldn't be used. The bottom line is that bioplastics need to perform and meet the needs of users in order to continue their growth—and it needs to be at a price point that doesn't disrupt a product or process.

Bioplastics Marketplace

Bioplastics Marketplace

> "As demonstrated in Europe, where countries and companies have embraced increased responsibility toward recycling and the management of waste, bioplastics will increase their emergence in the marketplace."

Bioplastics are not a new material, but that does not necessarily mean there is a mature marketplace for the material. With greater commercialization and marketplace presence, brands and consumers will better understand bioplastics. Further, as demonstrated in Europe, where countries and companies have embraced increased responsibility toward recycling and the management of waste, bioplastics will increase their emergence in the marketplace.

Estimating the Market Size of the U.S. Bioplastics Market

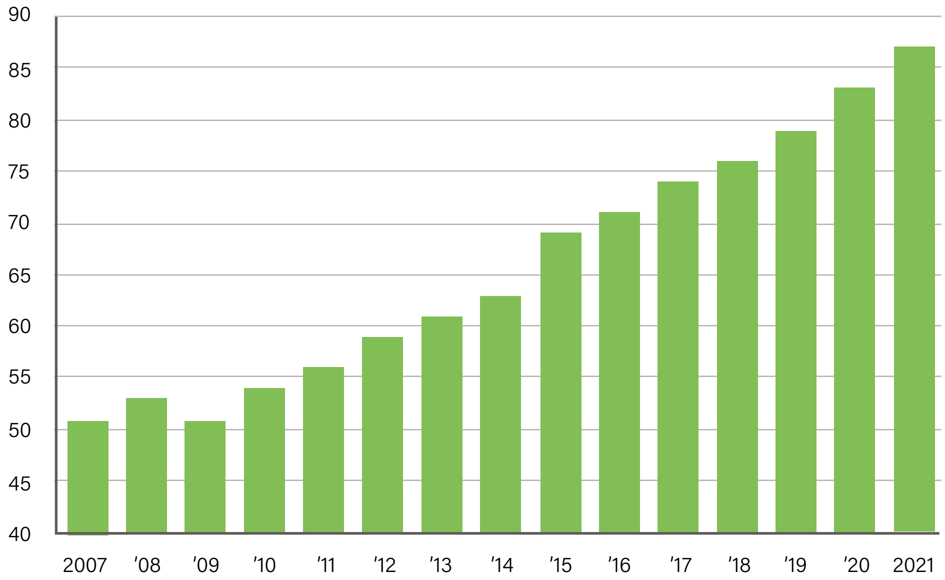
The life cycle of an industry can be summarized into four stages—introduction, growth, maturity, and decline. Pinpointing the market size of an industry in the growth cycle is the most challenging. While the U.S. plastics industry is a mature industry, bioplastics manufacturing industry is in the growth cycle phase—as modern bioplastics manufacturing did not take hold until the 1970s when oil prices rose by 400 percent. The lag between discovery of bioplastics to modern manufacturing in the 1970s is attributable to affordable petrochemical feedstocks.

IBISWorld estimates bioplastics manufacturing revenue in the U.S. at $177.9 million in 2016. Estimates on the global bioplastics market continue to vary—Dun and Bradstreet estimates the global market for bioplastics is forecast to grow from a value of $2 billion in 2014 to nearly $44 billion by 2020.

One of the defining characteristics of an industry in growth cycle stage is that it grows faster than the economy. With growth potential higher than other industries, bioplastics will continue to attract new entrants into the industry, increasing the competitive landscape. New entrants bringing in new products and manufacturing technologies make bioplastics manufacturing industry more dynamic relative to other industries. It is estimated that the number of establishments in bioplastics will increase by 70.5 percent from 2007 to 2021.

Bioplastics—Number of Establishments Estimate

Source: IBISWorld

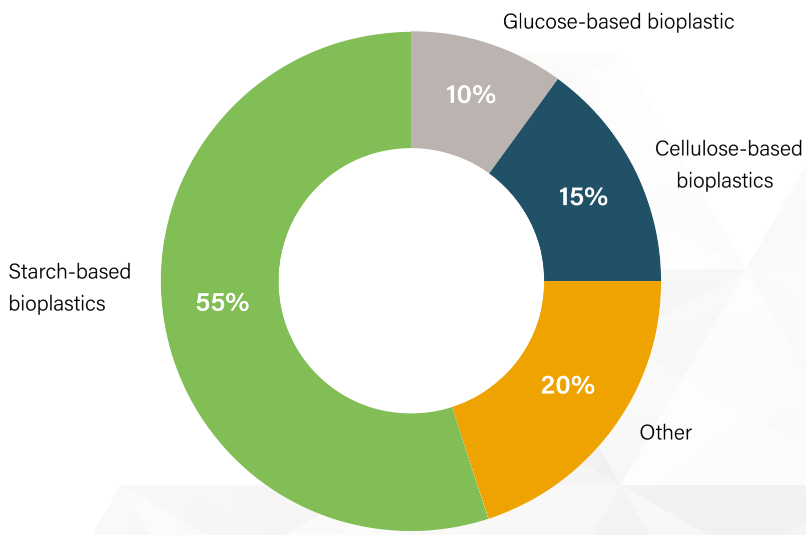


In addition to bioplastics manufacturing attractiveness to new entrants, growth industries are also known for their capital intensity. Product development is the key, using substantial amount of capital, putting research and development in a high gear that could lead to changes in the current product segmentation in bioplastics.

> "Product development is the key, using substantial amount of capital, putting research and development in a high gear that could lead to changes in the current product segmentation in bioplastics."

Product Segmentation (2016)

Source: IBISWorld

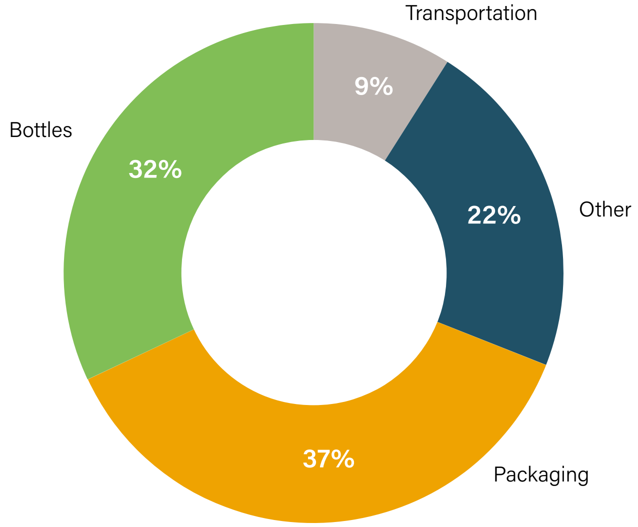


"The transportation market segment could increase—from single to double digit—as more bioplastics applications in transportation are explored and eventually used in the transportation industry space."

As bioplastics product applications continue to expand, the growth dynamics of the industry will continue shift. Looking at industry studies on market segmentation, packaging is the largest segment of the market at 37 percent followed by bottles at 32 percent. Other and transportation comprise the remainder of the market at 22 percent and 9 percent, respectively in 2016.

Major Market Segmentation (2016)

Source: IBISWorld



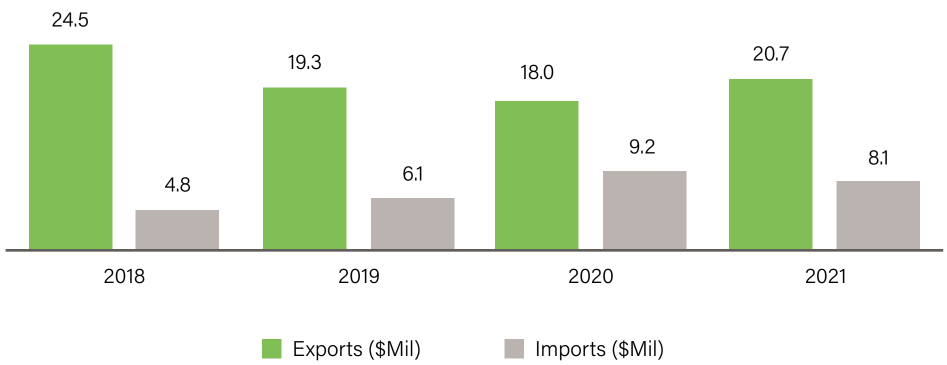
"Healthy levels of U.S. household consumption—due to low unemployment and rising disposable income—should result in stable demand in bottling and packaging. However, a shift between other and transportation segments could be expected," said PLASTICS chief economist Pineda. "The transportation market segment could increase—from single to double digit—as more bioplastics applications in transportation are explored and eventually used in the transportation industry space."

Will the market continue to grow?

In the plastics industry, the world is the market. However, bioplastics trade remains small relative to aggregate plastics industry trade. Imports normally satisfy less than 5 percent of domestic demand, while exports account for between 10 percent and 15 percent of revenue according to IBISWorld 2016 estimates. According to European Bioplastics, about 75 percent of biopolymer capacity is located outside the U.S. Trade estimates, benchmarked from 2016 dollars, in U.S. bioplastics are shown in the figure below. PLASTICS expects export numbers to stay positive and would come in higher than projections considering that the global economy has improved since the benchmark year with global economy expected to expand 3.9 percent in 2018 and 2019.

Bioplastics Trade

Source: IBISWorld

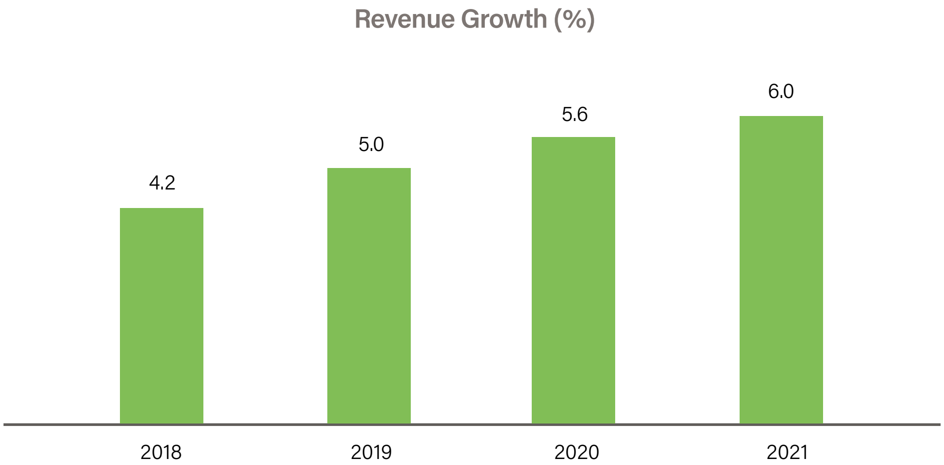


Growth opportunities in bioplastics manufacturing are expected to continue from the demand and supply side. While in the past growth in bioplastics was primarily driven by higher petrol-based polymers, changes in consumer behavior will be a significant factor for higher demand of bioplastics.

Revenue growth could be expected to come from both higher consumption of bioplastics ranging from food and beverage packaging, mulch film, carpet yarn face, filament for 3D printers, food serviceware, and others. Increasing consumer awareness of biobased products and packaging will sustain bioplastic demand growth. Based on PLASTICS survey, consumers expect to see more bioplastics in disposable cutlery and tableware, plastic bags, food packaging, and cosmetic packaging. (see Consumer Perceptions of Bioplastics: Page 27)

"Changes in U.S. tax policy, particularly the full expensing of capital expenditure, should support research and development in bioplastics. The overall low cost of energy in the U.S. complements nicely with research and development activities and manufacturing, which generates stable supply of innovative bioplastic products," Pineda added.

In the past two years, revenue growth rates were expected at 3.9 percent and 3.0 percent respectively. Industry research such as IBISWorld projected annual revenue growth for 2018-2021 at 4.9 percent. However, with more visible signs of U.S. and global economic growth, conservatively, annual revenue in bioplastics manufacturing could come in the 5.0 percent to 5.5 percent range over the same period.



If there is one thing that brand owners should keep in mind regarding bioplastics is that bioplastic packaging could be a strong product differentiator particularly to environmental conscious younger consumers—in addition to innovation to meet consumer needs—in sync with corporate sustainability initiatives.

Global Market Influences and New Bioplastic Formulations

> "Price of alternative raw materials for developing bioplastics will need to be at an acceptable price point for companies to utilize them."

Underlying the bioplastics marketplace, and its potential growth, is that traditional plastics manufacturers are tied to the oil and gas markets that can fluctuate in price and have a finite supply worldwide. According to the International Energy Agency, the plastics market growth will continue to increase the demand for petroleum. IEA analyst Tae-Yoon Kim told Bloomberg, "Petrochemicals will take center stage in driving oil demand. This is why oil majors are very much focusing on petrochemicals."

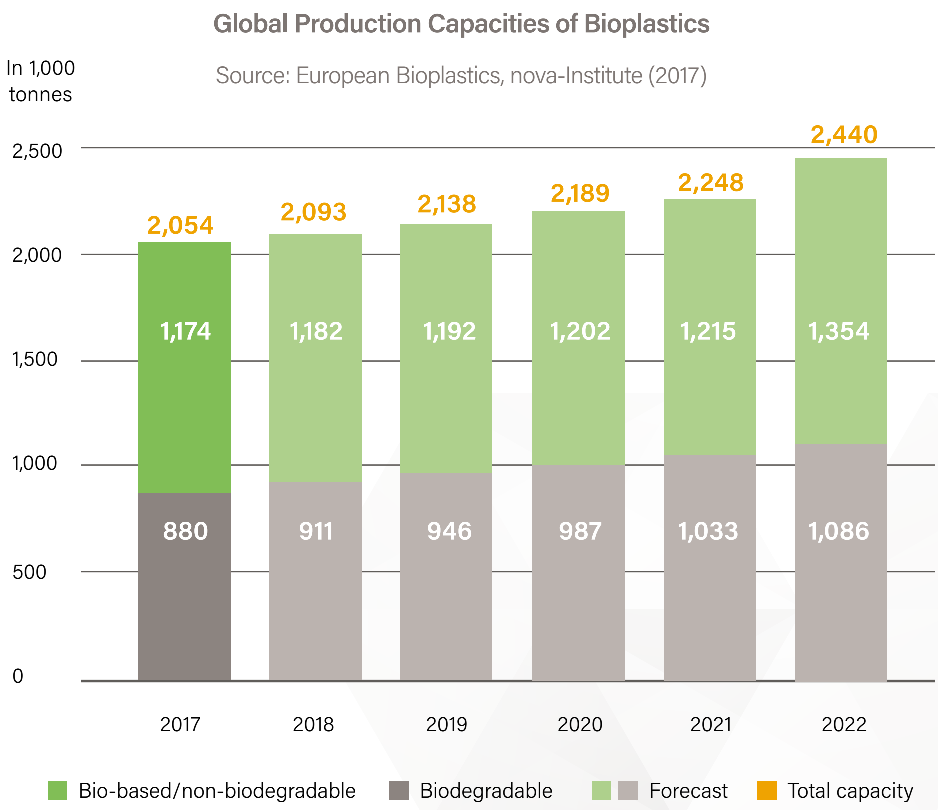
IEA has identified Saudi Arabian Oil Co., ExxonMobil Corp., Royal Dutch Shell Plc. and Total SA as major petrochemical companies working to expand their plastic footprints. Saudi Aramco projects petrochemicals to grow 4 percent annually.

The dynamic between plastics and bioplastics is likely to continue to intensify as measured by investments, research and development, brand adoption and consumer attitudes. But price of alternative raw materials for developing bioplastics will need to be at an acceptable price point for companies to utilize them. "Alternative raw materials must be competitive," Stora Enso's CFO stated to Bloomberg.

Additionally, the demand and utilization of bioplastics by major brands must expand. "It won't ever work if there's just one big consumer company like a Coca-Cola trying to drive suppliers," Ben Jordan, head of environmental policy at Coca-Cola told Bloomberg. "You need more demand out there in industry."

These changes and bioplastic adoptions among brands are taking hold. Novamont's Vice President, North America, Dan Martens said, "Beginning-of-life considerations are becoming more important as companies begin to measure their carbon footprints. Bottlers using conventional plastics are looking to lower carbon footprints by adding biomaterials, replacing fossil chemicals with plant based chemicals. It may or may not change the plastic, making it compostable for instance, but you gain sustainability and lower carbon footprint with the plant-based product."

According to EU Bioplastics, approximately 1 percent of the 320 million tonnes of plastics produced annually is classified as bioplastic—but that footprint continues to gradually grow. By 2022, the nova-Institute in Germany and EU Bioplastics predict bioplastics production capacity will climb from 2.05 million tonnes to 2.44 million tonnes.



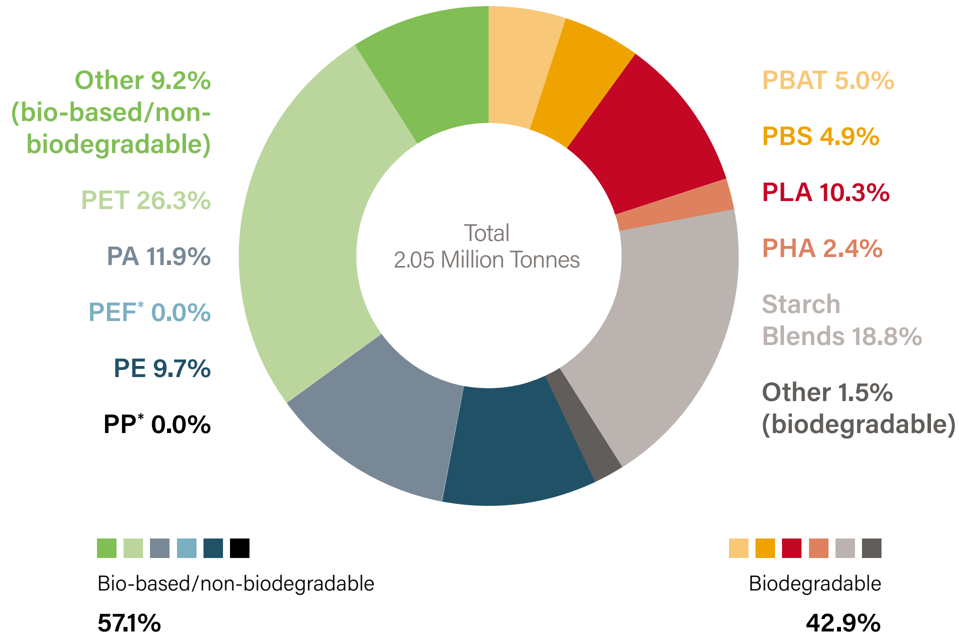
PET (polyethylene terephthalate), most commonly found in bottles and packaging (like the Coca-Cola PlantBottle), continues to be the most produced bioplastic around the world. But European Bioplastics predicts other polymers will take foothold in the marketplace and increase their usage—like PLA (polylactic acid) and PHA (polyhydroxy alkanoates) bioplastics.

PLA is produced with a variety of renewable resources, depending upon the location of its manufacturing. In North America, PLA is sourced to cornstarch while Asia utilizes cassava. Sugar cane is the source for PLA in many other areas. PLA is also increasingly being used in different applications, including sheeting for agriculture, injection molding applications and fabrics. Next generation formulations already underway are a high heat PLA that can reach melting points of 440 degrees Fahrenheit.

PHA polyesters are sourced to many microorganisms, and are 100 percent biobased and biodegradable and feature a number of properties appealing to customers. PHA production, European Bioplastics predicts, will triple in capacity during the next few years.

Global Production Capacities of Bioplastics 2017 (by material type)

Source: European Bioplastics, nova-Institute (2017)



*Bio-based PP and PEF are currently in development and predicted to be available in commercial scale in 2020,

Biobased PP (poly propylene) is also expected to expand in the coming years through injection formed applications in a number of industries including construction and automotive. PP also has film applications for food and medical applications. Biobased PUR (polyurethane) continues to have go-to applications in the marketplace and may even grow in the future compared to conventional PUR given the versatility of bio-sourced PUR. In 2016, European Bioplastics estimated approximately 40 percent of the global production capacity of bioplastics was PUR (polyurethane). For its 2017 calculations, however, PUR was not factored into the global capacity calculations, as there was no reliable data reporting the actual volumes of PUR containing bio-based content. ("The available data only shows how much biomass is used to produce bio-based polyols, which, however, is not sufficient in order to deduct reliable data on the actual volumes of bio-based polyurethanes," European Bioplastics noted.)

The Push for 100 Percent Biobased PET Bottle

The growth among some bioplastic formulations will have to compete with new PET formulations that expand the content of renewable sources. Most PET bottles currently have approximately 30 percent biobased material—a number of companies and collaborations are working toward the 100 percent biobased PET bottle.

Coca-Cola's PlantBottle, which will use sugar cane in all of its plastic bottles by 2020, is working with Virent to develop a 100 percent PET. Virent was among the first to develop a PET plastic bottle comprised entirely of plant materials in 2015, although on a demonstration scale.

Other companies—in the plastics business and major brands—are racing to develop a commercially scalable 100 percent PET plant-based bottle. That ultimate goal is getting close. Biofuels Digest's Jim Lane identified ten leading trendlines and initiatives chasing the completely renewable clear bioplastic bottle, "Challenges are abundant but the chemistry is powerful and the tipping-point may well arrive soon."

Anellotech, a New York-based sustainable technology company is working toward developing a 100 percent biobased PET plastic with partners Toyota and Suntory. Like Coca-Cola, Suntory currently has a 30 percent plant-based bottle for one of its brands, but wants to increase its environmentally friendly bottling position.

Anellotech's focus is on its Thermo Catalytic Biomass Conversion (Bio-TCat™) technology that uses renewable, non-food biomass (wood, sawdust, corn stover, sugar cane bagasse—a byproduct, and other non-food materials) that is rapidly heated, producing gases that are converted to hydrocarbons by the company's reusable catalyst.

Danone and Nestlé Waters are collaborating with Origin Materials, a California startup, to develop and launch at commercial scale a PET plastic bottle made from 100 percent biobased material. The effort, called NaturAll Bottle Alliance, is focusing on completely sustainable and renewable resources (recycled cardboard, sawdust, woodchips, rice hulls, etc.) to make a sustainable and renewable PET bottle.

Their collaborative effort, with the intent to offer the new bottle to all companies in the food and beverage space, is working toward a lightweight PET that is transparent, recyclable, and provides necessary protective qualities to products—while being better for the environment. Their focus is on renewable feedstocks that do not utilize resources or land needed for food production is critical.

> There is not enough bioplastics to replace all fossil based plastics, but the place for bioplastics is where people are tying to improve their carbon footprint with plant based material or where they support a food scrap collection composting system. We need to look for an added value or benefit for replacing a fossil based material—like compostable or plant based material.
>
> **—Dan Martens, Novamont**

PEF (polyethylene furanoate), a 100 percent biobased plastic innovation under development by a number of companies is expected to be trialed in the next few years. PEF has similar qualities to PET (strong barrier properties) and features that will be suitable for packaging food, drinks and similar products. The plastic would be biobased and recyclable—and useful for food packaging and bottling usages.

A priority for PEF is establishing a supply chain for FDCA, the chemical building block needed for the production of PEF. BASF is collaborating with Avantium to produce polyethylene furanoate FDCA—and PEF. The companies have worked with several major brands involved in bottling and packaging to test PEF to ensure its safety, its ability to fulfill the criteria of major brands, and demonstrate the recyclability of PEF bottles.

Tom van Aken, CEO of Avantium told Plastics News Europe that the companies were working to commercialize PEF with a wide range of potential market applications. On the recyclability of PEF, van Aken told PNE, "We want to work with the recycling community to really learn about the after-use of PEF, how it will be recycled and how it will enter the systems of recycling PET."

The companies are working to determine the percentage of PEF that can be added to the PET recycling stream, estimating it will be minimal and less than 10 percent. "You can add percentages of PEF into the PET stream without having an impact on the quality of the material, because PEF is chemically very similar to PET unlike the comparison with PLA and other materials," van Aken told Plastics News Europe.

The environmental debate on bioplastics

A debate developing among environmental and some non-governmental organizations is the usage of bioplastics and biodegradable plastics and whether they are better for the environment. *Environmental Leader* recently looked into this issue and reported: "It's a complicated issue."

Richard McKinlay, a resource recovery specialist at Axion, said individual brands and companies using plastics in their packaging must comprehensively evaluate their needs and make a decision upon the current waste management infrastructure and level of public understanding. McKinlay told Environmental Leader, "They also need to understand what actually happens to their materials at end-of-life and what their environmental impact could be."

Companies need to carefully evaluate their specific packaging needs and marry that up with the array of plastic materials and processes available to them. "A toolbox of various polymers makes it easier for producers to make the product that works best for them and their customers. We support further bioplastics development insofar that it provides the opportunity for wider selection and capabilities," PLASTICS' Krieger said.

Made from renewable feedstocks, rather than petroleum, bioplastics is a sound sustainable option for brands and consumers; the material is "not free of environmental impact and the carbon emissions associated with growing crops and converting these into the required chemicals needs to be taken into account," McKinlay noted.

Continued advancements in biodegradable or compostable bioplastics (and waste management) are expected to improve their environmental performance. Additionally, as more brands move to adopt bioplastics packaging, consumers will increasingly understand their characteristics and the recycling or composting of the materials.

Total Corbion's Derek Atkinson believes many consumers currently have little understanding not only of bioplastics—but plastics in general. With bioplastics, he said it was critical for the industry to be transparent and accurate in describing them. "International and national standards and certification schemes, focusing on biobased and compostable bioplastics play a key role to prevent any misunderstanding. In the future, more attention will need to be given to the correct recommendation for 'end-of-life' for the plastic products (recycle, compost, etc.)," he said.

An additional point made by McKinlay was the advancements of collection and recycling infrastructure. "Ultimately it has to be down to infrastructure investment, public education and behavioral changes," McKinlay told Environmental Leader. "Plastics are an inherent part of our lives and not 'all bad.' Their responsible use and disposal/recycling should be a top priority."

Martens of Novamont concluded, "As an industry, we need to take the responsibility and be an expert and consultant not only on our bioplastic products, but the whole landscape, the infrastructure in place, the goals of customer and the goals of community. For example, consumers looking 'to be more environmentally conscious' should consider whether moving from a PET bottle with an established recycling stream, to a bioplastic bottle that has limited recycling or local composting infrastructure capabilities could have a less overall environmental impact. We need to support sound initiatives and be clear with our bioplastics capabilities and currently available systemic benefits."

Bioplastics Facts vs. Myths

Bioplastics Facts vs. Myths

A number of participants in the plastics value chain are positioned to educate others on bioplastics—they have a responsibility to improve the understanding of them and the value they provide everyone. Unfortunately, there continue to be some myths about bioplastics in the marketplace today—and in the future. Keep these points in mind when you are discussing bioplastics:

Fact

Inventors have been tinkering with cellulose in plants since the 1800s. Parkesine was a biobased plastic developed in the 1860s. Maurice Lemoigne, a French scientist, made Polyhydroxybutyrate (PHB) in 1926. Given the abundance of petroleum at the time, bioplastics did not take hold in the marketplace, but by the 1940s and during World War II, bioplastics had a wide variety of uses and source materials, including soy, hemp, wood and other renewable materials. The modern era of bioplastics manufacturing started in the 1970s when oil prices skyrocketed.

Myth

All bioplastics are biodegradable. In fact, only some bioplastics are biodegradable. The ability for bioplastics to biodegrade varies—some bioplastics may be compostable (industrial or home) while others are marine biodegradable or anaerobically digestible. Even if they are biodegradable, the plastics must me managed properly at their end-of-life.

Fact

Bioplastics come in an array of different polymers—and uses. There are more than 21 different bioplastic polymers in the marketplace today or in final testing stages offering a wide variety of end uses. The variety of bioplastics means it can fulfill a wide range of needs.

Myth

Advancements in bioplastics will address and solve end-of-life issues with plastics. Reality is all plastics, including bioplastics, need to be properly managed and disposed; expanded recycling and composting infrastructure is necessary for all plastics—and all waste. Continued consumer education and the adoption of advanced recycling programs will best address end-of-life issues for waste and plastics.

Myth

Bioplastics use vital food sources. Despite overwhelming evidence and statistics, there continues to be a concern—or argument—that agricultural feedstocks for bioplastics would be better used to provide food for growing population around the world. According to European Bioplastics, only .82 million hectares were used to grow renewable feedstocks in 2017, representing less than 0.02 percent of the global agricultural area of 5 billion hectares. Bioplastics don't take away food for humans—in fact, they help produce, protect and preserve food for humans and reduce food waste.

Fact

Bioplastics offer plastic material performance that are biobased, sustainable and biodegradable. Bioplastics are plastics that compete in the marketplace for customers—but they will not entirely replace traditional, petroleum-based plastics. Bioplastics are complementary to existing fossil-based plastics and offer customers new options and alternatives based on societal and consumer trends and demands.

Consumer Perceptions of Bioplastics

Consumer Perceptions of Bioplastics

For consumers, their introduction and initial understanding of bioplastics may come from the brands and companies they trust turn to them for their packaging or products. The PlantBottle from Coca-Cola, on the shelves since 2009, is an example of an early, marketed application that influenced consumer understanding of bioplastics.

Companies embracing social responsibility, and recognizing good public relations as well as economic opportunity, are at the forefront of adopting bioplastics—and working to articulate the material's benefits as part of their corporate culture to consumers.

"When a brand utilizes more sustainable packaging, i.e. high recycled content or plant-based plastics, it is presented as integral to the brand promise," said Steve Davies of NatureWorks. "People don't buy packaging, they buy the brand experience of which packaging can make an important statement. Consumers become better educated through long-term use, familiarity, and practice. Brands play a big part in this process."

> "We see that more and more consumers are aware of climate change and the impact of plastics leaking into the environment. Informed and conscious consumers are leading the transition, especially in the more developed countries. In some countries, like Italy and France, all consumers are aware of and part of the transition to bioplastics: in these countries compostable shopping bags/vegetable bags are mandatory and hence have entered everyday life. The explanations printed on those bags has also helped to show the benefits of bioplastics."
>
> **—Derek Atkinson, Corbion**

PLASTICS Survey Reveals Benefit of Educating American Consumers on Bioplastics

Perc Pineda, PhD, Chief Economist, PLASTICS

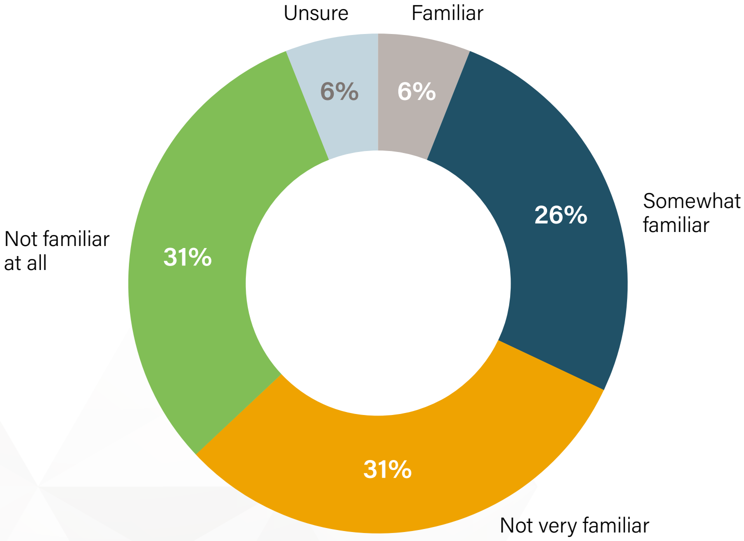
By Perc Pineda

The Plastics Industry Association surveyed 1,000 urban consumers in January-February 2018, ages 18 and above, to gauge their awareness of bioplastics. Survey results show that more than half are familiar about bioplastics and would prefer purchasing a product made from bioplastics. Further, consumers expect to see bioplastics in disposable plastic tableware, plastic bags, food and cosmetic packaging, and toys.

When asked how familiar they were about bioplastics, the net familiar response was 63 percent with 31 percent not familiar. (Of the consumers surveyed, 67 percent of those familiar with bioplastics were male and 59 percent were female.) However, only 38 percent believed they have used or seen a product made from bioplastics. More than a third—44 percent—were unsure if they have seen or used a bioplastic product.

Note: More than half of the survey respondents—52.5 percent—are in the $25,000–$125,000 household income range. The survey panel was almost virtually split in the middle in terms of gender—47.3 percent were male and 52.7 percent where female.

Consumers' Familiarity of Bioplastics



A solid 45 percent of consumers surveyed said bioplastics are more sustainable compared to the 5 percent who thought otherwise. Half of the consumers surveyed were unsure on bioplastics sustainability, which is consistent with the high percentage of respondents indicating to have not seen or used a bioplastics product.

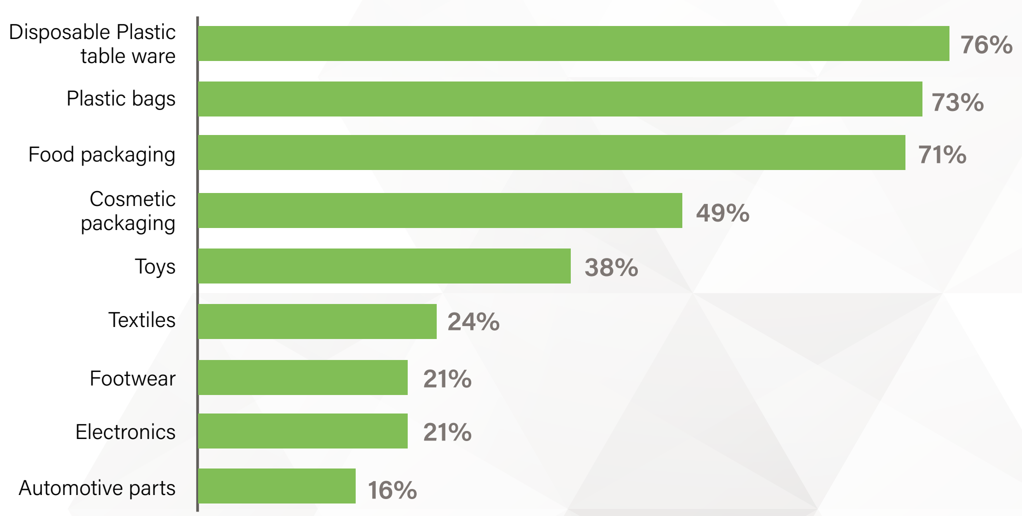
All other things being equal—including cost and performance—more than half of consumers surveyed (64 percent) would prefer to buy a product made from bioplastics, while only 5 percent of respondents indicated would not. However, the number of consumers willing to pay a premium for products made from bioplastics was lower—slightly above half (54 percent). Twenty four percent of consumers surveyed indicated unwillingness to pay higher prices for bioplastic products.

Of the consumers surveyed, 48 percent indicated that they were likely to recommend bioplastics product to others while 11 percent will not. This suggests that almost half of consumers of bioplastics are more likely to be promoters. Considering the 11 percent who would not recommend bioplastics yields 37 percent of consumers surveyed as net promoters of bioplastics.

More than half (55 percent) of consumers surveyed were more likely to buy a plastic product bearing the USDA BioPreferred label, while only 7 percent was not. It is important to note that 26 percent of consumers surveyed reported not having seen the government label.

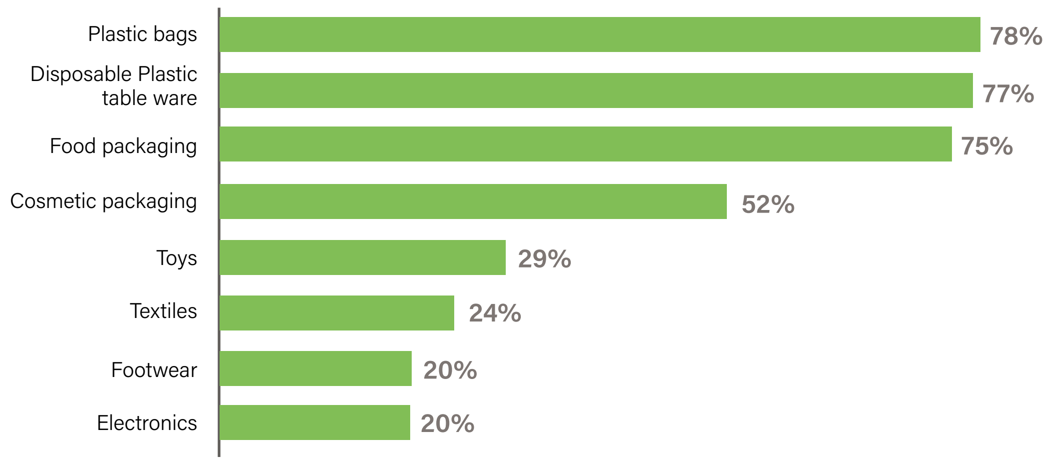
On the type of products consumers expect to see a biobased bioplastic—a plastic made wholly or in part from renewable materials such as sugar cane or cornstarch—disposable plastic tableware tops the list. A high 76 percent of consumers would expect disposable plastic tableware such as plates, forks, cutlery and straws to be made from biobased bioplastics. Plastic bags and food packaging follows the lead as shown in the chart below. On biodegradable bioplastics—a plastic that will completely breakdown through a natural process—78 percent of consumers surveyed expect to see disposable plastic bags made of such material. Also, a high 77 percent of consumers expected to see biodegradable bioplastics in disposable plastic dinnerware, and 75 percent in food packaging. Consumer expectations on products types made of biodegradable bioplastics are shown in chart on page 30.

Percentage of Consumers Expecting Biobased Bioplastic by Product Types



Both charts reveal that a high percentage of consumers expect to see bioplastics—whether bio based and or biodegradable—in the top five product types: disposable plastic tableware, plastic bags, food packaging, cosmetic packaging, and toys. Survey results show that performance ranks first when it comes to purchasing decisions on products using or made of plastics. Consumers purchase products and expect they will perform. By a narrow margin, consumers surveyed indicated price as the second consideration—in sync with a general view that consumers expect products to perform as priced. Whether the plastic is recyclable, sustainable, and biobased ranked third, fourth, and fifth, respectively in consumer purchasing decisions of products using or made of plastics.

Percentage of Consumers Expecting Biodegradable Bioplastic by Product Types



Emerging Markets for Bioplastics

Emerging Markets for Bioplastics

Although the PLASTICS survey indicated consumers do not expect to find bioplastics in the automotive sector, auto companies like Ford Motor Company have long histories with bioplastics—indicating consumers likely interact and use bioplastics without knowing it. As noted, the market growth of bioplastics will go well beyond bottles and packaging, in part, due to the wide breadth of different kinds of bioplastics. Bioplastics are increasingly being used for fibers for personal care products, like diapers, as well as infusion products such as tea bags, filters used in coffee capsules, and specialty medical care products. In addition, many 3D printing technologies rely upon bioplastic filaments.

Steve Davies, Director—Performance Packaging, NatureWorks LLC said, "Once we get beyond categorizing bioplastics as just biobased and/or biodegradable and confining bioplastics to packaging, we make real strides in terms of understanding where, when, and why these functional polymers provide the better solutions in terms of meeting business, environmental, and social needs."

Increasingly, as companies look at their environmental practices and carbon footprint, they are turning to bioplastics for help with green solutions. Bioplastic manufacturers and developers have the expertise to partner with companies on developing new products, packaging, and processes that utilize bioplastics.

Novamont's Dan Martens recommends that bioplastics companies collaborate extensively with their customers regardless of the application. "In transitioning to bioplastics, go slow and take small steps. You have to touch it and spend time with the material. While polyethylene is pretty much bulletproof, bioplastics are different and have unique, specific applications," Martens said. "When a customer moves toward a new bioplastic application, we always look at the specifications, take it to our labs and do practical trials. Then, we recommend using the new application in one store or a small trial. Let stakeholders become familiar with it and accustomed to it."

PLASTICS Innovation in Bioplastics Award

The Bioplastics Division of the Plastics Industry Association named DowDuPont Industrial Biosciences and Archer Daniels Midland as the winners of the 2017 Innovation in Bioplastics Award.

The two companies developed a method to produce furan dicarboxylic methyl ester (FDME), a biobased monomer, from fructose derived from cornstarch.

The new FDME-producing technology is more sustainable and results in higher yields and lower energy and capital expenditures than traditional conversion methods. Biobased FDME has the potential to replace petroleum-based materials in many applications with high performance, renewable materials in industries like packaging, textiles and plastics engineering.

The partnership unites two leaders on the material supply side and manufacturing side; they have been collaborating since 2016 on developing new polymers. One DowDupont official said, "They're experts in making sugar out of corn kernels and we're experts in doing some of the chemistry."

Search for New Breakthroughs and Market Applications

European Bioplastics anticipates bioplastics comprised of sugar cane, wood and corn will grow by 50 percent over the next five years paced by traditional industry players like BASF, Cargill Inc., and Mitsubishi Chemical Holdings—as well as new start-ups and companies diversifying. Cross-border partnerships and alliances between mature companies and start-ups are pushing the development new bioplastics and sustainable sourcing materials.

BASF is working with Scandinavian-based Stora Enso, a company traditionally focused on paper, to increase production of cornstarch-based plant bottles. The two companies see Coca-Cola's growing demand as a trend for other brands and applications; the two companies are developing a facility in Belgium to produce 50,000 tons of per year.

Stora Enso has been investing in bioplastics research and focusing on its core business as a source for its products. The company is currently testing a plastic bottle with 50 percent wood fiber.

DowDuPont scientists continue their search for new polymers and applications—including plastics developed from sugar. Paul Fagan, who heads up the company's research into sustainable polymers told the Wilmington News Journal, "A lot of our customers and, we ourselves, would like to get away from using oil to make things. We're not only trying to make things sustainable but also…recyclable at the same time, that's the advantage of the particular class of polymers I'm working on now."

Fagan noted that renewable materials from corn and sugar cane, produce lower carbon emissions than plastics developed using traditional petroleum processes, and the key building block to the materials under development are renewable.

DowDuPont already has plastics that partially utilize renewable materials, including its Sorona polymer-based material used in clothing that is 37 percent renewable plant-based and uses 30 percent less energy in its manufacturing processes compared to traditional Nylon 6 and releases 63 percent less greenhouse gas emissions. Sorona was among the first products designated by the USDA's BioPreferred Program for certified biobased products.

Eastman Chemical is also looking to collaborate outside the petrochemical industry to expand the use of its new fiber Naia; targeting the world of clothes designers and companies. Introduced in 2017, Naia is cellulosic yarn derived from renewable wood pulp sourced to sustainably managed forests—the manufacturing process is a near-closed-loop process where all waste is either recycled, reused or offered for resale.

And while wood pulp may not conjure up images of soft fabrics and comfort—that is exactly what Naia delivers with a hypoallergenic nature of the fabric. Eastman's Naia fabric offers softness, high breathability, and efficient moisture management; it is also certified by the BioPreferred program.

Although Naia was initially targeting the intimate apparel market—given the fabric's softness—it is also suited for the growing sports fashion sector. Gernoa B. Jacobs, global director, Textile Fibers at Eastman said, "We are very excited as the textile industry hasn't seen a new yarn for a while. You are getting a luxurious yarn without giving up on comfort, the emphasis is on luxury, comfort and function. Naia has huge versatility and is excellent for digital printing. The performance sector is a market we have our eye on."

University Research on Bioplastics

Bioplastic research is also being pursued at universities—often as part of a partnership with an industry player or through federal research grants. Iowa State University hosts the Center for Bioplastics and Biocomposites, a National Science Foundation research center. Year-to-year, Iowa State is pursuing a number of projects on bioplastics through NSF funding.

Stanford University and IBM have united on an effort to develop biodegradable plastics using plants that can be produced economically and at scale. "We have created catalysts that help accelerate reactions so we can make polymers faster, and the technology will enable us to create different polymers with different catalysts. For instance, we could create a polymer for high-density versus low-density polyethylene," IBM researcher Gavin Jones told Waste360.

One aspect of the joint effort by IBM and Stanford is to develop a bioplastic that can decompose in about 90 days if it is composted and within a year should it end up on a landfill.

Arizona State University researchers are using cyanobacteria, an organism that uses photosynthesis to create sugar, along with a second bacteria that consumes the sugar to produce a bioplastic. The cyanobacteria are captured in microscopic hydrogel beads produced from seaweed extract that are submerged in saltwater filled with bioplastic-producing bacteria.

According to lead researcher Taylor Weiss at ASU, "Trapping the cyanobacteria in a hydrogel is also critical—it means that the same cyanobacteria can be reused instead of regrown, and because the trapped cells barely grow at all, energy otherwise spent on growth can be redirected toward even greater sugar production."

University of Nebraska biological systems engineering professor Yiqi Yang is conducting research on polylactide (PLA), a biodegradable polymer sourced corn starch or sugar cane. The Nebraska team is looking to a new process that heats PLA to 400 degrees before slowly cooling the material; this process minimizes the need for chemical solvents and increases the materials resistance to heat and moisture. Yang believes PLA could be a valuable biodegradable textile fiber in that it meets the criteria used in dyed textiles.

To date, the team has only used this thermal process to test polylactide, but they believe other biobased materials could work as well.

"Yang believes PLA could be a valuable biodegradable textile fiber in that it meets the criteria used in dyed textiles."

Brand Innovations Using Bioplastics

Take a look at sectors and brands partnering with the bioplastics industry to diversify their products and introduce new uses for bioplastics: Add images of products below?

Mars

Developing bioplastics for specific brands' needs requires collaboration and time, but they do provide rewards. Candy manufacturer Mars started working with Rodenburg Biopolymers in 2012 to develop a biobased wrapper from potato starch waste for its Snickers bar. In 2016, the candy, wrapped in its new biobased film, was awarded the 2016 "Oskar" at the 11th annual Global Bioplastics Awards competition.

The new film did not impact the company's manufacturing processes or shelf life in any way. But the development did take extensive cooperation and commitment by all the companies involved in the development, manufacturing, printing and packaging. Thijs Rodenburg, CEO of the bioplastics company told Greener Packaging, "The unique cooperation is a best-practice example for the whole bioplastics industry. Without a joint effort, this success could not have been realized."

Aveda

Aveda, a skin, hair and beauty company, was among the first in their sector to use 100 percent post-consumer recycled PET. The company has embraced environmental sustainability—in their products, manufacturing processes, and packaging—as they work to appeal to consumers and differentiate their products. Aveda is also combining post-consumer recycled materials with bioplastics for some of its packaging needs. Its bioplastics are primarily sourced to sugar cane bagasse—the byproduct of sugar production

Reebok

Reebok shoes is partnering with DowDuPont Tate & Lyle Bio Products to develop an entirely compostable shoe. The Cotton + Corn initiative combines the best technologies and stylings of both brands. The bioplastics company is providing Reebok its Susterra® propanediol, a 100 percent certified biobased product derived from corn. When combined with organic cotton, the Reebok shoe will be petroleum-free, non-toxic and help fulfill the company's philosophy and sustainability efforts.

Bill McInnis, Head of Reebok Future said, "This is really just the first step for us. With Cotton + Corn we're focused on all three phases of the product lifecycle. First, with product development we're using materials that grow and can be replenished, rather than the petroleum-based materials commonly used today. Second, when the product hits the market we know our consumers don't want to sacrifice on how sneakers look and perform. Finally, we care about what happens to the shoes when people are done with them. So we've focused on plant-based materials such as corn and cotton at the beginning, and compostability in the end."

McInnis, who says Reebok is "growing shoes," wants to develop a compostable biobased shoe in the future.

Lego

The building blocks of children around the world since the 1950s—Lego—are now turning to bioplastics. The Danish-based company with the interlocking bricks announced their iconic pieces will begin to be made with plant-derived materials as part of their sustainability efforts. The first pieces going-green will be parts representing leaves, bushes and trees—all to be made with polyethylene made with ethanol from sugar cane.

This change is "a great first step in our ambitious commitment of making all Lego bricks using sustainable materials," according to Tim Brooks, Lego VP of environmental responsibility. Lego has pledged to transition to using sustainable materials in Lego products and packaging by 2030.

McDonald's

Canadian McDonald's in 2017 introduced 100 percent compostable McCafé® Keurig®-compatible pods for their coffee products. Certified by the Biodegradable Products Institute, the pods are made using plant-based materials such as coffee bean skin. The effort will help reduce waste generated at the restaurants as the pods use technology that allows it to break down in five weeks.

Naturepedic

Bioplastics companies sharing their technologies and environmental advantages with brands also led to Braskem, Sealed Air and Naturepedic to partner in developing a renewable polyethylene foam for mattresses. All the companies involved in the partnership have roots in environmentally friendly technology and practices.

Braskem's I'm Green Polyethylene (PE) is a biobased resin from ethanol, a renewable resource the company obtains from Brazilian sugar cane; the material has a negative carbon footprint in that sugar cane utilizes C02 and releases O2.

"We have been watching the industry moving more and more toward renewable materials and noticed a great deal of energy being dedicated specifically to bioplastics," said Chad Stephens, VP of Research and Development for Sealed Air. "For Sealed Air, the next logical strategic step was to progress the advancement of renewable polyethylene foams in our product offerings, enabling our clients to further reduce their packaging footprint with more sustainable and enviro-friendly solutions while still providing superior protection against damage."

The first application of the renewable polyethylene foam will be in Naturepedic's mattresses.

Biome Bioplastics—Coffee Shop Products

Coffee shops are testing Biome Bioplastics' new disposable coffee cups, lids and pods specifically designed with natural and renewable resources including plant starches and tree by-products like cellulose to be fully compostable and recyclable—while matching the heat and stress performance offered by petroleum-based plastic.

Biome offers a wide range of bioplastics to different industry sectors, but is advancing its coffee bioplastics to address a specific need of coffee shops, restaurants and take out shops.

Paul Mines, CEO of Biome told British Plastics & Rubber, "For such a simple product, disposing of a single coffee cup is a very complex problem. Coffee drinkers are acting in good faith when they see recycling logos on their takeaway coffees but most cups are lined with oil-based plastic and the lids made of polystyrene making recycling impossible, even when placed in the right bin. Our solution is making biopolymers that can be made into fully biodegradable coffee cup and lid combinations. The result being a biobased takeaway cup disposable either in a paper recycling stream or food waste stream."

The cups and lids can compost into carbon dioxide and water with three months in appropriate conditions, Mines said.

Automotive Sector's Continued Embrace of Bioplastics

The automotive sector has had a leadership role in bioplastics and sustainability practices in its products and manufacturing. Car companies around the world have expanded their usage of plastics and bioplastics to save weight and improve mileage performance. Car manufacturers are increasing their use of recycled materials, developing plant-based renewable parts, eliminating substances of concern, addressing end-of-life impacts with improved recyclability, and partnering with environmentally-focused suppliers.

Ford Motor Company

Ford was among the first companies to embrace soy-based structural plastics in 1941 and hemp-based plastics in 1942. More recently it has used soy-based PUR foams for seat components, wheat-straw filled plastics, and cellulose for interior body applications. According to Alper Kiziltas, a member of Ford Motor Company's Research and Innovation Center, all of the company's vehicles in North America source their seat backs, seat cushions, head rests, and some headliners to soybean oil-based bioplastics.

The average Ford vehicle uses 20 to 40 pounds of renewable materials, including soy-based foams, seals and gaskets, castor-oil based PA 11 (Polyamide 11) foams, and natural fiber reinforced plastics.

In using the renewable materials, the company is strict in its rules regarding safety and quality standards. Renewable materials must meet appearance requirements (weather, stability, etc.), odor and VOC (volatile organic compound) issues, performance requirements (stiffness and impact strength, moisture sensitivity, chemical compatibility,) global sourcing availability and cost concerns.

Ford is continuing its push to improve bioplastics technology with its materials. Torrefaction (roasting of wood or other biomass to remove moisture to create a product that has increased energy density, is easy to handle and transport) is seen by the company as being a compatible sourcing material with thermoplastics. The process is being used to develop new PP (polypropylene) composite headlamp housing that replaces talc with better performance and less weight.

"The advantage with switching out fiberglass with agave is the agave is more recyclable. The fiber doesn't break when remolding, so there is potential there to reuse it in the same product."

Using waste from the manufacturing of tequila, Ford is also working to use the fibrous byproduct of the spirit to develop a new bioplastic for its vehicles. Ford will use agave fibers to manufacture plastic components in upcoming vehicle models (Fusion and Lincoln MKZ), viewing the fibers as a strong, resilient renewable resource.

David Grewell, an associate professor at Iowa State University, indicated many companies are looking at agave as the fibers are flexible yet retain their strength.

"The advantage with switching out fiberglass with agave is the agave is more recyclable. The fiber doesn't break when remolding, so there is potential there to reuse it in the same product," Debbie Mielewski, Ford's senior technical leader of sustainable products, told the Guardian.

Ford is also using the Coca-Cola PlantBottle technology as part of an interior fabric in its Fusion Energi hybrid vehicle under development. The company's 2018 Fusion was given the Society of Plastics Engineers Automotive Division Innovation Award for Environmental for its next generation sustainable content biofoam that the company collaborated on with International Automotive Components and BASF.

Around the world, Ford is looking to available renewable resources to supply its bioplastics needs, including algae in North America, eucalyptus in South America, bamboo in Asia-Pacific, and hybrid poplar in Europe. The aim of these efforts is to identify a widely available, regional source material at a lower cost—as well as the possibility for planting in non-agricultural lands.

Ford, according to Kiziltas, is even looking at plastic ocean waste as a source material. He told Plastic News, "We need to make sure we work together—engineers, scientists, suppliers, to find a solution for this idea. And we are looking to the oceans. Can we make car parts using ocean plastics?"

Mazda

Mazda launched its collaborative (industry-academia-government) bioplastic initiative in 2008 focusing on non-food-based cellulosic biomass with the intention of using the material in its vehicles beginning in 2013. The company wanted to use inedible vegetation such as plant waste and wood shavings so the material would be carbon neutral, but also widely available.

At the launch of the initiative, Seita Kanai, Mazda's director and senior executive officer in charge of R&D, summarized the program, "Development of a non-food-based bioplastic made from sustainable plant resources has great potential in the fight against global warming, and can help allay global food supply concerns. Mazda is pleased to join forces with our regional partners as we work toward systematically combining various biomass technologies."

At the Tokyo EcoPro 2016 trade show on green technologies, Mazda rolled out its Roadster RF convertible coated in bioplastic. Manufactured with Mitsubishi Chemical Corporation, the Roadster RF utilized the new bio-engineered plastic from polymer isosorbide which does not require exterior paint and is also scratch and weather resistant.

Mazda had already been using the bioplastic in the interior of vehicles, including the Axela sedan, Demio, and CX-9 SUV.

Isosorbide has been used in medicine successfully for years—its contribution for bioplastics has proven to be a game-changer for Mazda. The company reported, "This key component is added to PET to produce PEIT. The addition of isosorbide means that it has better heat resistance and better optical clarity to the plastic. This will enable the company to produce parts that are as durable as conventional painted ABS plastic parts, yet feature a higher-quality finish and the associated design advantages. And since the bioplastic can be dyed and does not require painting, it also reduces emissions of volatile organic compounds. Dying the material gives the parts a deep hue and smooth, mirror-like finish of a higher quality than can be achieved with a traditional painted plastic."

Renault

France-based Renault is also using the isosorbide-based plastic, Durabio. The auto company is using the bioplastic as an outer mask of the new Clio vehicle's speedometer-tachometer dashboard. With the material's transparency, it highly colorable with dyes and does not require painting—it has better impact strength and weatherability compared to 100 percent petroleum-derived plastics.

Renault, doesn't advertise or promote its bioplastics usage, but does use the material to replace heavier materials in its vehicles in the search of better fuel mileage, has a keen eye on the cost of materials.

Toyota

Toyota has embraced bioplastics for specific parts and components for a number of years—part of the company's drive to improve mileage performance and its environmental footprint. Toyota uses bioplastic to make seat cushions and is collaborating with the Society of Automotive Engineers' (SAE) International Green Technology Systems Group to develop an industry guide on sustainability for the automotive sector.

Toyota is using plant-based plastics developed by Denso in its car navigation systems, according to the automotive supplier. Denso, a global automotive component supplier, is providing Toyota a bio-polycarbonate (PC) made from starch and urethane resin extracted from castor oil. The material provides necessary hardness for use in vehicles while also being very malleable in developing shapes for complex designs—like bezels in vehicle dashboards.

Denso is also providing biobased plastics components that are used under the hood in vehicles. In 2009, Denso and DowDuPont developed a castor oil-based radiator tank that is more environmentally sustainable.

Public Policy Impact on Bioplastics

Public Policy Impact on Bioplastics

Public policy changes on a wide range of issues frequently come down to policies that provide incentives (a carrot) or penalties (the stick). Bioplastics and the management of plastics are no different in the U.S. and around the world. Government policies and funding can influence the development and adoption of bioplastics—whether mandates on recycling, composting and waste reductions or through research funding to advance the development of new materials.

Europe, which has a lead in bioplastics utilization, is also at the forefront of shifting consumer behaviors via public policy that reflect the carrot and stick principle. Analyst group FMI ties the growth of bioplastics usage in the EU to regulatory schemes that support bio-applications as well as consumer awareness and acceptance of green products and policies.

"The EU and some individual countries in the EU are quite far advanced, whereas some others are still exploring the benefits bioplastics bring as part of a circular economy. The Japanese government has tasked the ministries to come up with a comprehensive strategy to stimulate bioplastics and reduce CO2 emissions," commented Derek Atkinson of Total Corbion PLA.

EU countries have been shifting consumer usage toward compostable or anaerobically digestible plastics—and particularly bags—for a few years.

The move stems from a mandate that put compostable bags at the forefront of consumer shopping habits.

In Italy, the mandate is currently being implemented to ban single use plastic bags for grocery shopping and the handling of fruit, vegetables and baked goods in favor for industrially compostable bags. These bags—found throughout the interior of stores near produce, bakery and meat sections—come with a fee of 1 to 3 euros per bag in Italy. EU nations have long embraced reusable mesh bags for carry groceries, but Italy's implementation and roll-out of the industrial compostable fruit and vegetable bags has been met with some consumer disagreement, although it is estimated that the annual cost per family would only be from $4 to $12.50 euro per year.

FKuR, a German-based bioplastics manufacturer with a range of biobased and biodegradable resins, has been focusing its product development and technology toward meeting EU-member country requirements. FKuR's Bio-Flex polymer is a home compostable, tear resistant product that is 40 percent biobased that is being used for fruit and vegetable bags. In 2017, FKuR introduced its new biobased TPE (thermoplastic elastomers) called Terrapene® that can use up to 80 percent renewable resources depending upon the grade.

In early 2018, the EU ambassadors approved new provisional waste reduction agreements on targets for recycling and reduction of landfilling. According to European Bioplastics, the new agreement is a boost for bioplastics as it "acknowledges that biobased feedstock for plastic packaging as well as compostable plastics for separate biowaste collection contribute to more efficient waste management and help to reduce the impacts of plastic packaging on the environment." Under the new directive, by 2023, separate collection of biowaste is set to be mandatory throughout Europe.

In the U.S., much of the federal government's efforts could be more characterized as carrots toward the use and adoption of bioplastics technology. Local governments, have to date, been more aligned with sticks to shape behaviors.

"Those jurisdictions that have been the most successful have employed both the carrot and the stick in a coherent way at the national level. The fragmented manner of U.S. regulatory systems at the city, state and federal levels are likely to lead to pockets of excellence rather than national solutions," Total Corbion's Atkinson said in an interview.

As mentioned previously, PLASTICS believes there is no one perfect polymer. While bans and other material deselection efforts may support another segment of the industry, they ultimately reduce consumer choice and access to the materials that may best meet their functional needs.

The U.S. government funds an array of research efforts to advance the development of bioplastics. A number of agencies and departments partner and support research being conducted at universities and labs around the country.

One major avenue of government-funded research is the National Science Foundation and its Center for Bioplastics and Biocomposites operating at Iowa State University and Washington State University. More than 20 businesses and public-private partnerships are also involved in the Center for Bioplastics and Biocomposites research activities and funding. In recent years, the Center has funded research projects aligned with identified priority areas: synthesis and compounding, biocomposites, processing, biobased products, modeling, and commercialization.

Another carrot/stick incentive, now more than 15 years old, is the U.S. Department of Agriculture's BioPreferred program, the leading federal government initiative to increase the purchase and use of biobased products. The program, which many in the industry believe is a benchmark, mandates purchasing requirements for federal agencies and contractors and sets a voluntary labeling initiative for biobased products and brands.

For purchases, the program has identified 97 different categories of products (cleaners, carpets, lubricants, paints) with biobased qualities. Each product has minimum biobased content requirements. At its inception, the program had only six product categories.

As part of the 2018 Farm Bill, the USDA is reviewing the program for potential reforms. For its part, the USDA sees significant benefits in the program; their 2016 Economic Impact Analysis of the U.S. Biobased Products Industry concluded that market-developing federal purchases have fostered growth in the bioproducts sector generating $393 billion in value added and supported 4.2 million jobs in 2014.

The diversity of federal agencies supporting—or looking at—bioplastics is exemplified by the U.S. Department of Defense (DoD). While maybe not immediately identified as a "green" agency, DoD is continually looking for ways to address sustainability issues. For the DoD, sustainability is a matter of national security.

The DoD's Strategic Environmental Research and Development Program/ Environmental Security Technology Certification Program has noted "the logistics of managing packaging waste in remote areas and foreign countries is becoming increasingly problematic and costly for the military."

The DoD believes polyhydroxyalkanoates (PHA), can functionally replace over 50 percent of their plastics used today. PHAs provide "properties from strong, moldable thermoplastics to highly elastic materials to soft, sticky compositions, and can be made as resins or as latex with excellent film-forming characteristics. PHAs are also biodegradable in aquatic, soil, and municipal waste treatment environments, and can be composted."

The DoD is working on a project to demonstrate the use of PHA natural plastic for foamed packaging and stretch/shrink film applications by its branches. Specific PHA formulations were designed for each application, including foamed PHAs, shrink-wrap and films.

The agency's research team successfully demonstrated PHA could be formulated into both foam and film grades. The DoD noted the material could "improve logistics and reduce the costs of transporting and disposing of packaging waste that are burdening military operations worldwide through a substantial reduction in nondegradable plastic waste."

The Department also recognized PHA plastics would benefit the environment by reducing greenhouse gas emissions and lowering energy utilization for resin production relative to conventional fossil fuel-based thermoplastics.

In November 2017, DoD issued a proposal seeking the design and development of biodegradable bullets for use at Army training facilities. Not only were the bullets to be biodegradable, designers were also being tasked with the charge of developing a material that would sprout plants as part of their breakdown.

Army training ranges are littered with spent ammunition rounds and casings that are difficult to retrieve. The training rounds do not need to be lethal, opening the door for alternative materials to be used by soldiers. The DoD is requesting proposals for training ammunition that would be a plant-derived bioplastic that would be embedded with seeds that could regenerate the ground on the ranges. The U.S. Army Corps of Engineers has already successfully trialed a biodegradable round capable of generating seedlings.

A Tale of Two Stadiums

While the Seattle Mariners and Pittsburgh Pirates play baseball with slightly different rules, they are more alike in the way they, like many arenas and venues, are moving aggressively toward the use of bioplastics and composting.

But just like the designated hitter rule, there is another difference between the two ball clubs: the city of Seattle mandated that the Mariners (and other residents and businesses) move forward with composting while the Pirates have a composting initiative based upon the private sector and the economics of producing and selling high-grade compost.

Public policy can shape how businesses operate—but change can also be achieved by the private sector when like-minded interests align and the economics make sense. According to many observers, the U.S. is at the forefront of composting at stadia where closed loop systems are utilized.

Safeco Field

Major League Baseball gave its "Green Glove Award" to the Seattle Mariners in 2017 for the team's recycling efforts. The San Francisco Giants had a stranglehold on the award for nine straight years, but the Mariners, after putting 96 percent of its waste materials toward recycling, won the Green Glove last year.

To reach its league-leading diversion rate, the Mariners have used 100 percent compostable tableware over the past four years to help work toward a zero-waste goal.

"Nearly everything used at Safeco Field is recyclable or compostable, including food service items like plates, cups, straws, bottles, and utensils," the MLB Green Glove announcement said. "Compost and recycling bins have replaced garbage cans on concourses, and cleaning crews separate plastics and compostable waste by hand after each game."

PNC Park

Since the 2009 season, the Pittsburgh Pirates have been working with a local organics composting company, AgRecycle, Inc., to implement a stadium recycling and composting program.

Carla Castagnero, president of AgRecycle told BioCycle.Net that the Pirates were aligned with her business model after five minutes of their first meeting. "Usually they (corporations that are interested in initiating composting programs) want to do it because composting is philosophically a good thing to do and it meshes with their sustainability goals," Castagnero described. "It usually takes several meetings to evolve from a conceptual idea to an action plan. That was not the case with the Pirates. They were serious and jumped directly into discussing the operational details of the program, not the philosophy. They wanted to know how to get it done."

The program, branded "Let's Go Bucs. Let's Go Green," achieved a 28 percent diversion rate in its first year, by 2012, the ballpark and its patrons were diverting nearly 65 percent of its waste. AgRecycle can only accept source separated organic waste.

Castagnero has said the success has been a partnership with the team and other vendors inside PNC Park—compostable products are found throughout the ballpark. "Ballparks are so complicated because they have private suites, clubs, different restaurants, concessions and the stands," Castagnero said. "It's not that there is just one location that prepares and serves the food. Composting has to be something that the team owners want, and in turn, the team coordinates the staff and practices internally."

Conclusion

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Bioplastics are increasingly being used in industry sectors beyond bottling and packaging: automotive, agriculture, textiles, catering/food service, electronics, building and construction are all turning to bioplastics to address a wide range of needs. Bioplastics' footprint in the marketplace continues to grow as companies inside and outside of the plastic sector look to leverage the material for current and future uses.

It is the diversity and flexibility of bioplastics that allows brands and industry sectors to turn to them for solutions; but cost, performance, supply, and beginning-of-life/end-of-life factors are still part of the overall equation as to how brands and sectors utilize them. The understanding of bioplastics in the plastics industry is strengthening—and now bioplastics manufacturers are partnering with brands and others to establish solutions to packaging, manufacturing and recycling that are needed to sustain the growth.

According to European Bioplastics, 2.05 million tonnes of bioplastics were produced in 2016—58 percent were used for packaging followed by 11 percent for textiles. No other industry market sector was in double digits. However, Future Market Insights (FMI) predicts that automotive will be the second largest industry sector for bioplastics in 2020 with 27.5 percent of the market.

Europe—for a number of factors—has been at the forefront of developing bioplastics with respect to research and utilization. However, in terms of total production, European Bioplastics reports Asia is the leader with more than half the global bioplastics produced in 2017 (56 percent). Europe produced 18 percent of the bioplastics and North America produced 16 percent.

Beyond the capabilities of bioplastics, other factors are contributing to their growth. As indicated by the PLASTICS 2018 survey, consumers expect some products to be bioplastics, like disposable plastic tableware. In other circumstances, the move toward bioplastics is part of a company's overall strategy and commitment to environmental positioning and curbing carbon footprints. And finally, the adoption of bioplastics are taking hold as result of public policy shifts toward circular economic principles and the reduction of waste.

The bioplastic sector acknowledges confusion among consumers continues—and even the brands that they work with to implement bioplastics into their products. To address the misunderstandings, the bioplastics industry recognizes the need to fully partner with their customers and provide solutions that go beyond bioplastics, but also addresses beginning- and end-of-life issues.

Even with anticipated growth in bioplastics, the land usage for feedstocks are expected to remain very low as new innovations come forward and increased competition develops between different renewable feedstocks. Despite the market growth predicted in bioplastics over the coming years, the land use share for bioplastics will remain around 0.02 percent. Further, significant research is being conducted on renewable, non-food feedstocks that can regionally source bioplastics based upon local resources.

The education on bioplastics also extends to consumers, who are increasingly aware of bioplastics in the U.S. according to PLASTICS' survey, but have some misconceptions about the material. If the European experience holds true in the U.S., consumers will increasingly understand bioplastics as they interact with them in stores and in products. A wide range of new products and applications are being developed today with bioplastic components—the innovations will be environmentally sustainable while matching the performance of traditional plastic technologies.

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Plastics Market Watch Snapshot

Plastics Market Watch Snapshot

Confidence Level

Extremely High

High

Moderately Optimistic

Neutral

Moderately Pessimistic

Low

Extremely Low

EQUIPMENT OUTLOOK

Market conditions should remain strong in 2016, but the pace of growth in the shipments data is expected to decelerate after six years of strong expansion.

New technologies in blow molding and other manufacturing processes are reducing energy costs and improving performance.

RESIN PROJECTIONS

Lower oil pricing since Q4 2014 has reduced overall price levels of plastics, helping keep polymers competitive relative to other packaging materials like paper, tin, and glass.

Resins prices in 2016 are well below the levels from the same time last year, and processors in all segments of the plastics industry are passing much of these lower costs through to their customers.

Some brand owners have made public commitments to using recycled content, so may continue to use it even if the cost advantage of recycled plastics is gone.

Recyclers have often enjoyed a cost advantage to prime material, but with prime pricing so low in recent years, that advantage has all but disappeared.

PROCESSOR OUTLOOK

Bottling consistently ranks near the top of SPI's data on the industry's output with only packaging film and sheet (except packaging) being higher in terms of employment numbers, value of industrial shipments, and capital expenditures.

Blow molders environmental strategies include lightweighting, the use of bioplastics, increased usage of recycled content and design for recycling.

BRAND OWNER INITIATIVES

Brand owners and bottlers are working to improve sustainability by designing bottles with an eye toward increasing recyclability. Additional efforts by large retailers also increasing inclusion of recycled content in their store products.

Soda consumption is decreasing in the United States, however, water consumption has been increasing; both products are predominately packaged in plastic bottles.

The pace of innovation in the plastics industry is remarkable as multiple avenues are underway to explore opportunities to improve on issues like shelf life and flavor performance.

Innovation on front end of plastics and the development of new polymers is outpacing the back end, end-of-life management and recycling of plastics.

RECYCLING ADVANCEMENT

In the U.S., single-stream collection of household plastic recyclables is growing and participation in recycling is increasing.

92 percent of the U.S. population has programs available for PET bottles/jugs & jar recycling and aluminum beverage can recycling.

A recent national survey reflects American's self-reporting of recycling plastics is high–and consumers value the importance of recycling.

According to ACC/APR study, one of the largest barriers to plastic bottle recycling is that consumers continue to be unaware of the value plastic bottles represent after their usage.

Municipalities need to understand they can benefit from the sale of bales of bottles, including revenue sharing to fund educational programs and other costs of collection.

According to the Recycling Partnership, only an estimated 21% of U.S. population lives in communities providing a uniform collection program to residents of all building types.

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